the Journal of Strength and Conditioning Research™

OPEN

A 24-Week Combined Resistance and Balance Training Program Improves Physical Function in Older Adults: A Randomized Controlled Trial

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Abstract

Jiang, G, Tan, X, Zou, J, and Wu, X. A 24-week combined resistance and balance training program improves physical function in older adults: a randomized controlled trial. *J Strength Cond Res* 39(1): e62–e69, 2025—This randomized controlled trial assessed the feasibility of older adults performing combined resistance and balance training (RBT) and compared the effects of RBT on physical function with those of resistance training (RT) alone and with no exercise training in older adults. In total, 65 community-dwelling adults aged 60–74 years were randomly assigned to an RT ($n = 22$), RBT ($n = 22$), or control ($n = 21$) group. The exercise intervention consisted of RT or RBT for 45 minutes, 3 times per week for 24 weeks. The control group engaged in no exercise training. The compliance rates were 93% in the RT group and 92% in the RBT group. No training-related adverse effect was observed. In the RT and RBT groups, dynamic balance ($p = 0.017$; $p < 0.001$, respectively), maximum walking speed ($p = 0.014$; $p < 0.001$), upper extremity ($p = 0.013$; $p = 0.009$) and lower extremity ($p = 0.009$; $p < 0.001$) muscle strength, and spirometry ($p = 0.043$; $p = 0.018$) were significantly improved at 24 weeks compared with the control group. Only the RBT group showed significant improvement in normal walking speed ($p = 0.002$). Compared with the RT group, the RBT group showed additional significant improvements in dynamic balance ($p = 0.001$) and lower limb muscle strength ($p = 0.027$). The findings of this randomized controlled trial indicated that RBT was safe and feasible for older adults. Long-term RBT had better effects than either no exercise training or RT alone on improving physical function in older adults. Compared with RT, RBT additionally benefited dynamic balance and lower limb muscle function, which are clinically important factors in preventing falls, frailty, disability, and other geriatric syndromes.

Key Words: aging, resistance training, walking speed, muscle strength, resistance training

Introduction

Among the numerous changes that come with aging, degradation of physical function is particularly concerning. Physical function is a core element for the independence of older adults, and participating in social activities is crucial (7,29,32). Maintaining physical function can prevent a series of adverse health outcomes, such as falls, disability, cognitive impairment, nursing home admission, and death (39). Such prevention is important for individuals and their families, the community, and society in reducing the cost of aging. Thus, preventive interventions should be encouraged among older adults to promote beneficial changes in physical function.

Resistance training (RT) is an effective intervention for improving some aspects of physical function in older adults (17,30). Various RT modalities have been shown to improve muscle strength and power (5,31,43). However, age-related decline in physical function can only be partially explained by loss of muscle mass, muscle strength, and muscle power (17). Other fundamental aspects of motor control strongly affect performance in activities of daily living among older adults, including deterioration of dynamic balance and motor coordination (41). In recent years, increasing attention has been devoted to the use of combined resistance and balance training (RBT).

Resistance and balance training is a compound training method that involves RT on a substable support surface. This training provides a powerful proprioceptive stimulus to the body, and progressive overload may enhance neuromuscular adaptations (4). Such training is more task-specific and realistic than general RT alone or balance training alone for performing complex tasks in activities of daily living (19–21,33). A meta-analysis by Behm et al. showed that RBT improved strength, power, and balance in older adults (4). However, the comparisons assessed in most of the included studies were with nonexercising controls. More recently, studies have compared the effects of RBT with RT alone on muscular strength, power, and balance in older adults (8,14). Resistance and balance training has advantages in improving balance, functional mobility, reducing the rate of lower extremity injuries (3), and decreasing risks of falls in older adults. Additional advantages of RBT over RT for other aspects of physical function, such as walking speed (the sixth vital sign), (18) are unknown. Decreased walking speed is common in older adults and is also an independent risk factor for morbidity, hospitalization, disability, and mortality (34). To refine age-appropriate exercise prescriptions, a study that directly compares the effects of

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Journal of Strength and Conditioning Research 39(1)/e62–e69

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long-term RBT and RT programs on multifaceted changes in physical function in older adults may provide useful information for researchers, healthcare providers, and strength and conditioning professionals. In particular, determining whether RBT is more effective than RT alone for older adults in specific aspects of physical function is important. The results of such studies may expand the use of RBT in the elderly population.

The aim of the present study was to investigate the effects of long-term RBT on physical function in older adults. We hypothesized that both 24-week RBT and RT would improve physical function in older adults but that RBT would have additional benefits on certain aspects of physical function in older adults.

Methods

Experimental Approach to the Problem

A sample size of 18 subjects per group was calculated using G*power statistical software at $\alpha = 0.05$, $\beta = 0.9$, and 2-sided test conditions (28). To account for dropouts, the target sample size for each group in this study was 22. The study used a three-arm study design in which subjects were randomized equally to RT, RBT, or control groups. The subjects were coded using Excel software, and then the formula "=RAND ()" was used to generate the corresponding randomized sequence. The subjects were sorted and grouped according to the randomized sequence. Randomization was performed by professional information technology staff to ensure that the investigators were not aware of subject recruitment and grouping.

Subjects

The study was conducted from February 2023 to September 2023 in Shanghai, China. Subjects were recruited through self-promotion, posters, and community outreach. Inclusion criteria were adults aged 60–74 years who were cognitively normal and able to live independently in the community, had no regular exercise habits, and volunteered to participate in this study by completing an informed consent form. Exclusion criteria were diseases or conditions that would interfere with physical function testing and training and uncontrolled hypertension (systolic blood pressure >160 mm Hg and diastolic blood pressure >100 mm Hg) (27). Subject characteristics are shown in Table 1. At baseline, there were no statistically significant differences in the assessed characteristics between the groups. As indicated in the guidelines outlined in the Declaration of Helsinki, all risks associated with the experimental procedure were explained before subjects provided written informed consent. The study was approved by the Ethics Committee of Shanghai University of Sport (No. 102772022RT123) and was registered at www.chictr.org.cn (ChiCTR2200056090).

Procedures

Testing consisted of 2 time points, baseline (pretraining) and post-training (after 24 weeks). Testing in the control group consisted of the same time points and intervals as the RT or RBT group, and the control group maintained their original lifestyle of sporadic, occasional exercise.

Resistance and Balance Training and Resistance Training Intervention. Before the start of training, subjects completed 2–3 familiarization sessions on a vertical vibration platform. The exercise program for older adults in the experimental groups included structured RT or RBT interventions, which were performed on Mondays, Wednesdays, and Fridays for 45 minutes each day across a total of 24 weeks. Subjects trained in groups of 5–6 people, and there were 4 groups in total. During the intervention, if subjects were unable to come to the training for any reason, individual "supplementary training" sessions were organized. Subjects rested for 2–3 minutes between different movements (17), and the rest time between each set is shown in Table 3. Training loads were incrementally increased and adjusted according to the International Strength and Conditioning Association (17) and the Expert Consensus Guidelines for International Exercise Recommendations in Older Adults (26). The RT group performed moderate RT on a vibrating platform (without the perturbation function turned on). The specific training protocol is shown in Table 2. The RBT group stood on a vibrating platform with the perturbation function turned on to perform RT. Different levels of instability were provided to the RBT group by adjusting the disturbance gears of the vibrating platform (10 gears in total, gears 1–7 and A1, A2, and A3), as detailed in Table 3.

The exercise intensity for both RT and RBT group interventions was moderate as assessed using a combination of subjective and objective indicators. Our findings confirmed that the use of the subjective rating of perceived exertion was valid for monitoring training load, with good reliability and internal consistency (16,22). Subjects were asked about their perception of exercise exertion after each training set. Rating of perceived exertion scale is scored out of 20 with a moderate intensity between 11 and 13 (40). OMNI-Resistance Exercise Scale is scored out of 10 with the same intensity between 4 and 5 (38). Exercise intensity was objectively monitored using a Polar heart rate monitor (Polar OH1; Polar Electro Oy, Kempele, Finland) and maintained at a 40–59% of heart rate reserve for moderate intensity (36). For safety, a paramedic-qualified fitness instructor, community health worker, healthcare provider, or physician supervised and closely monitored subjects for chest tightness, shortness of breath, respiratory distress, injuries, and other conditions during exercise training. Reactions of the subjects were observed after exercise and medical attention was sought in cases of serious adverse feelings. To motivate the subjects to adhere to the program and to help them to develop a lifelong habit of adhering to evidence-

Table 1
Subject characteristics.*

Variable	Control group (n = 21)	RT group (n = 22)	RBT group (n = 22)	F	p
Age (y)	69.43 ± 8.07	69.23 ± 5.51	67.27 ± 6.78	0.659	0.521
Height (m)	1.64 ± 0.07	1.61 ± 0.08	1.60 ± 0.07	1.613	0.208
Body mass (kg)	61.12 ± 8.74	61.67 ± 9.08	61.00 ± 9.03	0.035	0.965
BMI ($kg \cdot m^{-2}$)	22.86 ± 2.50	23.91 ± 3.99	23.89 ± 2.74	0.780	0.463

*RT = resistance only training; RBT = combined resistance and balance training; BMI = body mass index.

Table 2
Progression in the resistance training intervention program.

Training movement	Weeks of exercise intervention					
	1–2	3–7	8–12	13–17	18–22	23–24
Warm-up	10 min	10 min	10 min	10 min	10 min	10 min
Standing heel raises	2 × 8-12	2 × 10-15	2 × 10-15	3 × 10-15	3 × 10-20	3 × 15-20
Chair sitting and standing	2 × 8–12 (47 cm)	2 × 10–15 (47 cm)	2 × 10–15 (43 cm)	3 × 10–15 (43 cm)	3 × 10–20 (40 cm)	3 × 15–20 (40 cm)
Half squat	2 × 30 s	2 × 30–60 s	2 × 30–60 s	3 × 30–60 s	3 × 30–90 s	3 × 60–120 s
High step in place	2 × 30 s	2 × 30–60 s	2 × 30–60 s	3 × 30–60 s	3 × 30–90 s	3 × 60–120 s
Squat with bicep dumbbell curl (women/men 2.5/4.0 kg)	2 × 8-12	2 × 10-15	2 × 10-15	2 × 10-15	2 × 10-20	2 × 15-20
Half squat lateral walk	2 × 8-12	2 × 10-15	2 × 10-15	3 × 10-15	3 × 10-20	3 × 15-20
In-place lunge	2 × 8-12	2 × 10-15	2 × 10-15	3 × 10-15	3 × 10-20	3 × 15-20
Relaxed stretching	5 min	5 min	5 min	5 min	5 min	5 min

based exercise, this study rewarded the subjects with certain incentives.

Measures. The results of a previous study by our team determined that the relevant indicators of physical function in community-dwelling adults aged 60–74 years include usual walking speed (UWS), maximum walking speed (MWS), the timed up and go (TUG) test, 30-second arm curl test (ACT), 30-second chair stand test (CST), handgrip strength, and spirometry (28). The intra-group correlation coefficients for the above indicators of physical function measured in this study were 0.91–0.99. Testers were blinded to group assignments at the pretest and posttest time points.

Anthropometrics. Height (meters) and body mass (kilograms) were measured using a standard stadiometer and a scale, respectively. The body mass index was calculated as the mass in kilograms divided by the square of the height in meters.

Thirty-Second Arm Curl Test. The ACT is a general measure of upper body strength, which is important for performing many everyday activities. The test involved counting the number of times a subject could curl a hand weight, 2.3 kg for women and 3.6 kg for men, through the full range of motion in 30 seconds.

Thirty-Second Chair Stand Test. The CST is a measure of lower body strength, which is an important aspect of fitness in older adults because of its role in common everyday activities. The test involved counting the number of times, within a 30-second period, that a subject could come to a full stand from a seated position with arms folded across the chest.

Timed Up and Go Test. The TUG test is used to assess agility and dynamic balance, which are important in tasks that require quick maneuvering. The test assessed the number of seconds required for a subject to stand from a seated position, walk 3 meters, turn, and return to the seated position.

Walking Speed Tests. The walking speed tests included UWS and MWS measurements. In the tests, a subject walked more than 10 m from the starting position to the end point at both their usual and maximal walking speed (as fast as possible without running). The walking time was recorded with a stopwatch for 6 m of walking between 2 to 8 m to avoid the effect of acceleration in the first 2 m and deceleration by braking in the last 2 m on speed (26).

Handgrip Strength Test. The handgrip strength test is a common measure of upper body strength. The maximum strength of a subject's dominant hand was tested using an electronic handgrip strength meter (Model: 4601a; Takei Kiki Kogyo Co., Niigata, Japan). The test was performed twice, and the best performance was recorded.

Spirometry Test. Spirometry tests primarily reflect the potential capacity of the respiratory function of the subject. A spirometry tester (model: GMCS; Beijing Xindonghua Tengfei Sports Equipment Co., Ltd., Beijing, China) was used for measurement. During the test, subjects stood in a natural position holding the handle of the spirometer in their hand and inhaling deeply until they could no longer inhale. The subjects then exhaled slowly into a mouthpiece until they could no longer exhale. Each subject was tested twice, and the maximum values were recorded. The results were accurate to 1 ml.

Statistical Analyses

SPSS 26.0 software was used to statistically analyze the data. Descriptive statistics were calculated for the demographic variables and basic information indicators of the subjects. Data that conformed to a normal distribution were analyzed using one-way

Table 3
Speed of action, perturbation gear, and rest times at different phases of the RBT intervention program.*

Period	Speed of movement (duration of centripetal and centrifugal contraction)	Scrambling gears	Rest time between sets
1–2 weeks	3 s; 3 s	1–2	2–2.5 min
3–7 weeks	3 s; 3 s	3–4	1.5–2.0 min
8–12 weeks	2 s; 4 s	5–6	1.0–1.5 min
13–17 weeks	3 s; 3 s	A1	0.5–1.0 min
18–22 weeks	As fast as possible; 3 s	A2	1.0–1.5 min
23–24 weeks	As fast as possible; 3 s	A3	1.5–2.0 min

*RBT = combined resistance and balance training.

analysis of variance (ANOVA) to compare differences between groups. For data that did not fit a normal distribution, the Kruskal-Wallis method was used for analysis. Changes in test metrics in the RT, RBT, and control groups before and after the interventions were analyzed using repeated-measures ANOVA, and Bonferroni's post hoc test was used to assess the interaction between the main effects of time and group. The results are expressed as mean and *SD*s (M ± *SD*). The level of significance was set at $\alpha = 0.05$, with *p*-values ≤0.05 considered statistically significant. Changes for all variables within a group were calculated using the formula $\Delta\% = ([M_{pre}/M_{post}] - 1) \times 100$. To improve readability and homogeneity, effect sizes were calculated before and after the intervention for each group. The thresholds for effect size were considered trivial (0.0–0.19), small (0.2–0.59), moderate (0.6–1.1), large (1.2–1.9), and very large (2.0) using the recommendations of Hopkins et al. (25).

Results

Dropouts and Attendance

Figure 1 shows subject recruitment and group assignments for the entire trial. In total, 78 older adults who met the inclusion criteria were recruited. Four of them did not sign the informed consent form, and 8 of them did not have the time required to complete the training. Thus, 66 people participated in this study. They were randomly assigned to the RT, RBT, and control groups, with 22 individuals in each group.

A total of 65 subjects completed the study. One older adult in the control group was excluded at week 7 of the study because of also participating in regular exercise. Subjects in the RT and RBT groups completed 67 and 66 of the 72 training sessions, respectively, within 24 weeks, with compliance rates of 93 and 92%, respectively. No adverse events were observed in either intervention group during the experimental period.

Between-Group Tests

At baseline, there were no differences in physical function tests between the groups ($p > 0.05$). After the interventions, the between-group tests for the changes in the TUG test and CST results were significant ($p < 0.05$). The post hoc tests showed that the values for the RT group ($p < 0.05$) and the RBT ($p < 0.05$) were higher than those for the control group and that values for the RBT group were higher than those for the RT group ($p < 0.05$; Figure 2). The between-group tests for the changes in spirometry, MWS, and ACT were also statistically significant ($p < 0.05$). The post hoc tests showed that the values for the RT group and for the RBT group were higher than those for the control group (both $p < 0.05$). The between-group test for changes in UWS was statistically significant ($p < 0.05$), with post hoc tests indicating that the value for the RBT group was higher than that for the control group ($p < 0.01$).The between-group test for the change in handgrip strength after the interventions was not significant ($p > 0.05$).

Within-Group Tests

Table 4 reports the mean ± *SD* and effect sizes for the pretraining versus posttraining measures of the physical function indicators. As training progressed, all physical function test indices improved in both the RT alone (3–17%) and RBT (5–32%) groups (Figure 2). In the control group, performance on the CST decreased (−6%), but no other indicators of physical function changed significantly.

Discussion

The findings of this randomized controlled trial indicated that RBT and RT alone interventions performed for 45 minutes 3 times a week for 24 weeks were safe and feasible for older adults, with no training-related adverse events occurring during the interventions. The adherence rates were higher than the 80% threshold (24). Both long-term RBT and RT alone improved physical function in older adults, which was consistent with our hypothesis. Compared with RT alone, RBT additionally benefited dynamic balance and lower limb muscle function, which are clinically important factors in preventing falls, frailty, disability, and other geriatric syndromes.

There was a significant improvement in the dynamic balance of the older adults in the RBT group compared with the control group, with a magnitude consistent with what other studies have found (8,9,37,42). The improvement in the TUG test for the RBT

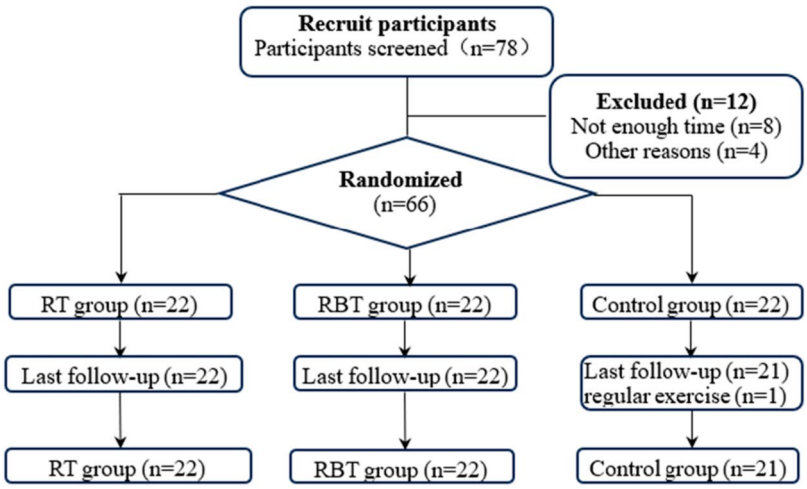

Figure 1. Flow diagram of study subjects in the resistance training (RT) only, combined resistance and balance training (RBT), and control groups.

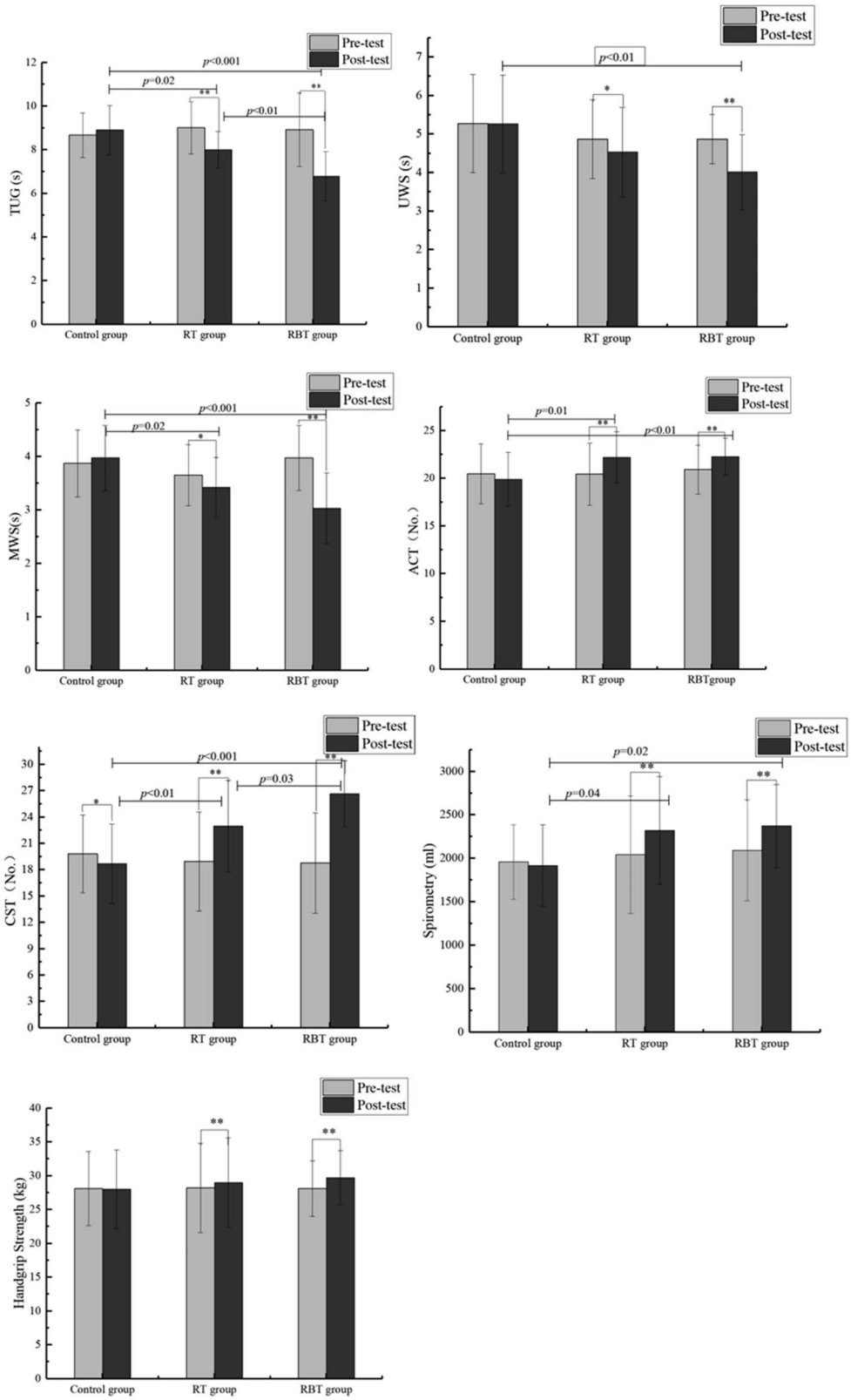

Figure 2. Changes in physical function within and between groups. TUG, timed up and go; UWS, usual walking speed; MWS = maximum walking speed; ACT = 30-second arm curl test; CST = 30-second chair stand test. *Statistically significant effect of $p < 0.05$; **$p < 0.01$.

group was greater than that for the RT group, which was consistent with our hypothesis. Resistance training can improve overall body muscle function (15). Resistance and balance training can activate the synergistic benefits between the locomotor and stabilizer muscles, enhance the activity of the neuromuscular system, and not only improve the overall function of

Table 4

Effects of the 2 exercise interventions on physical functions in older adults.*

	CON (n = 21), mean ± SD (95% CI)	p	ES	RT group, mean ± SD (95% CI)	p	ES	RBT group, mean ± SD (95% CI)	p	ES
TUG (s)									
Pre	8.67 ± 1.03 (8.20 ± 9.14)	0.14	0.04	9.01 ± 1.20 (8.48 ± 9.54)	<0.001	0.42	8.92 ± 1.68 (8.17 ± 9.66)	<0.001	0.76
Post	8.90 ± 1.14 (8.39 ± 9.42)			8.00 ± 0.83 (7.63 ± 8.36)			6.78 ± 1.12 (6.28 ± 7.28)		
UWS (s)									
Pre	5.27 ± 1.27 (4.69 ± 5.85)	0.94	<0.001	4.87 ± 1.02 (4.42 ± 5.32)	0.02	0.09	4.87 ± 0.64 (4.59 ± 5.15)	<0.001	0.39
Post	5.26 ± 1.27 (4.68 ± 5.83)			4.53 ± 1.16 (4.01 ± 5.05)			4.01 ± 0.97 (3.58 ± 4.44)		
MWS (s)									
Pre	3.87 ± 0.63 (3.58 ± 4.16)	0.30	0.01	3.65 ± 0.57 (3.40 ± 3.90)	0.04	0.07	3.67 ± 0.51 (3.45 ± 3.90)	<0.001	0.35
Post	3.97 ± 0.61 (3.69 ± 4.24)			3.42 ± 0.56 (3.17 ± 3.67)			3.03 ± 0.66 (2.73 ± 3.32)		
ACT (no.)									
Pre	20.48 ± 3.26 (19.05 ± 21.91)	0.06	0.06	20.45 ± 3.26 (19.01 ± 21.90)	<0.001	0.37	20.91 ± 2.56 (19.77 ± 22.04)	<0.001	0.27
Post	19.90 ± 2.84 (18.61 ± 21.20)			22.18 ± 2.70 (20.98 ± 23.38)			22.27 ± 1.96 (21.41 ± 23.14)		
CST (no.)									
Pre	19.81 ± 4.43 (17.79 ± 21.83)	0.02	0.09	18.95 ± 5.66 (16.44 ± 21.46)	<0.001	0.55	18.77 ± 5.75 (16.22 ± 21.32)	<0.001	0.83
Post	18.67 ± 4.55 (16.59 ± 20.74)			22.95 ± 5.21 (20.65 ± 25.26)			26.64 ± 3.75 (24.97 ± 28.30)		
Spirometry (ml)									
Pre	1957.90 ± 429.32 (1762.48 ± 2,153.33)	0.17	0.52	2041.41 ± 674.65 (1742.29 ± 2,340.53)	<0.001	0.59	2091.36 ± 584.25 (1832.32 ± 2,350.41)	<0.001	0.60
Post	1916.33 ± 470.27 (1702.27 ± 2,130.40)			2,320.55 ± 620.66 (2045.36 ± 2,595.73)			2,373.32 ± 476.32 (2,162.13 ± 2,584.51)		
HS (kg)									
Pre	28.14 ± 5.54 (25.62 ± 30.67)	0.37	0.01	28.21 ± 6.56 (25.30 ± 31.12)	<0.001	0.31	28.12 ± 4.07 (26.32 ± 29.93)	<0.001	0.62
Post	28.01 ± 5.81 (25.36 ± 30.65)			29.00 ± 6.61 (26.07 ± 31.93)			29.65 ± 4.04 (27.86 ± 31.44)		

*CI = confidence interval; ES, effect size; RT = resistance only training; RBT = combined resistance and balance training; TUG = timed up and go; UWS = usual walking speed; MWS = maximum walking speed; ACT = 30-second arm curl test; CST = 30-second chair stand test; HS = handgrip strength.

body muscle but also mobilize the small muscle groups in the trunk that play a stabilizing function (6). There were marked improvements in the TUG indexes in the RT and RBT groups, with the improvement in the RBT group showing greater clinical significance than that in the RT group, which was also consistent with our study hypothesis. These findings have important implications for healthy aging. The higher degree of instability in the RBT may have induced additional active balance adaptations (14). Concurrent balance and RT initiate synergistic effects in the nervous system. All these factors may be related to the greater improvements in lower limb muscle strength and dynamic balance in the RBT group than in the RT group observed in this study. However, it has also been shown that force, power, and speed of movement are compromised when performing RBT compared to RT (1,2).

Both RBT and RT were clinically important in this study for improving walking speeds in older adults. A change in walking speed of 0.05 $m \cdot s^{-1}$ is the recommended standard for a clinically meaningful change (35). In this study, the UWS increased by 0.09 $m \cdot s^{-1}$ in the RT group and 0.27 $m \cdot s^{-1}$ in the RBT group, which was consistent with the results of previous studies (8,21). The MWS increased by 0.11 $m \cdot s^{-1}$ in the RT group and by 0.37 $m \cdot s^{-1}$ in the RBT group after 24 weeks of intervention. Thus, all increases in walking speed far exceeded the minimum of 0.05 $m \cdot s^{-1}$. Increased pace is a strong predictor of survival, with a reported 58% reduction in the relative risk of death (13,23). Maximum walking speed places greater demands than UWS on the balance control system, and it requires higher levels of conscious control and cortical activity (12). Thus, MWS provides more comprehensive information on physical function than UWS (11). The results of this study expanded the meaningful impact of RBT interventions on MWS.

Compared with the control group, the RBT and RT groups showed comparable improvements in upper extremity strength, which may be related to the inclusion of dumbbell arm curls in the training program. Resistance training and RBT did not elicit meaningful improvements in handgrip strength compared with the control group. Hand grasping is a highly evolved task with a particularly complex neurophysiological basis (10). A study by Eckardt (14) reported no meaningful improvement in handgrip strength in healthy older adults following lower extremity RBT. This is similar to our study where there was no statistically significant change in handgrip strength following RBT despite a trend toward improvement compared to the control group. Both the RT group (12%) and the RBT group (12%) had clinically significant and comparable increases in spirometry. This finding provides supporting evidence for the benefits of moderate-intensity RT and RBT interventions on spirometry function, while expanding the value of RBT in promoting pulmonary ventilation function.

This study has limitations. We did not monitor dynamic changes in the indicators of physical function in the middle of the intervention, preventing short-term or longitudinal comparisons. The subjects were not blinded to the study group allocation. In addition, the subjects in this study were 60–74 years of age without functional limitations, limiting the extrapolation of the findings to adults beyond this age range and to adults in this age range with functional limitations.

The findings of this randomized controlled trial indicated that long-term RBT performed 3 times a week for 45 minutes was a safe, feasible, and effective training method to improve physical function in older adults. Compared with RT alone, RBT had additional benefits on dynamic balance and lower limb muscle function, which are important for delaying functional dependence and reducing the burden of care. The findings of this study also provided supporting evidence that for older adults, RBT had a better effect and efficiency in increasing the body's functional reserve. The study findings provide a reference for the selection of evidence-based exercise intervention programs for older adults.

Practical Applications

To improve physical function and reduce functional dependence in later life, coaches and conditioning professionals can implement RT alone or RBT combined for 45 minutes, 3 times per week, at a moderate intensity for better long-term outcomes in older adults. To prevent falls, weakness, and disability in older adults, coaches and healthcare providers can provide supervised combined RBT, which may be a more beneficial mode of exercise than RT for improving dynamic balance and lower extremity muscle strength.

Acknowledgments

The authors are grateful to Shangti Health Technology (Shanghai) Co., Ltd., for their support in recruiting subjects and to X. Wu's research group for their assistance in data collection. No funding was provided for the current work. G. Jiang conceived and designed the experiments, analyzed the data, prepared the figures and/or tables, authored or reviewed drafts of the article, and approved the final draft. X. Tan and J. Zou performed the experiments, prepared figures and/or tables, and approved the final draft. X. Wu conceived and designed the experiments, authored, and approved the final draft. The authors report no conflict of interest. This study was supported by The Program for Overseas High-Level Talents at Shanghai Institutions of Higher Learning (No. TP2020063) and Ministry of Education Humanities and Social Sciences Program (21YJC890053).

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